

CHRIS MISQUITH

[LinkedIn](#)

[GitHub](#)

Email: misquith.chris@gmail.com

Mobile: +91 9731470993

ABOUT ME

I'm an Artificial Intelligence student with a strong interest in Python programming and Data Structures & Algorithms in C++. I'm curious about computer hardware and how it all works together with software. I try to stay consistent with my work, enjoy team projects, and like learning new things through building and experimenting.

EDUCATION

Sardar Vallabhbhai National Institute of Technology

B. TECH in Artificial Intelligence; CGPA: 6.81

Surat, India

August 2024 – Current

Gulf Indian High School

CBSE Grade 12: 82.2%

CBSE Grade 10: 82.4%

Dubai, UAE

July 2018 – April 2024

SKILLS SUMMARY

- **Languages:** Python, SQL, JavaScript, HTML, Kotlin, C++
- **Frameworks:** Pandas, NumPy, Matplotlib
- **Tools:** Excel, PowerPoint, MySQL
- **Platforms:** Jupyter Notebook, Android Studio, Visual Studio Code
- **Soft Skills:** Rapport Building, Excellent communication, Team Collaboration, Time Management

PROJECTS

ScreenCalibrator (Screen Calibration Android App) | [LINK](#)

Engineered a system-wide screen overlay application using modern Kotlin, Jetpack Compose, and Coroutines, allowing users to dynamically adjust display brightness, RGB gains, and colorblindness filters.

- **Architected a Unidirectional Data Flow (UDF)** using **StateFlow** to synchronize a Material 3 control panel with a persistent background **ForegroundService**, ensuring zero-latency visual updates.
- **Overcame Android's SurfaceFlinger security sandbox constraints:** Since true GPU-level **ColorMatrix** manipulation requires Root/Superuser access, I developed a custom "alpha-compositing" engine. This engine mathematically blends dynamic RGB layers over the WindowManager to successfully simulate/imitate hue rotation, saturation drops, and contrast compression for non-rooted devices.
- **Managed strict Android 13/14+ OS integrations**, handling runtime overlay permissions (SYSTEM_ALERT_WINDOW) and implementing compliance for **FOREGROUND_SERVICE_SPECIAL_USE** policies.
- **Tech Stack:** Kotlin, Jetpack Compose, Coroutines, StateFlow, WindowManager API, Android Graphics (Canvas/Color).